

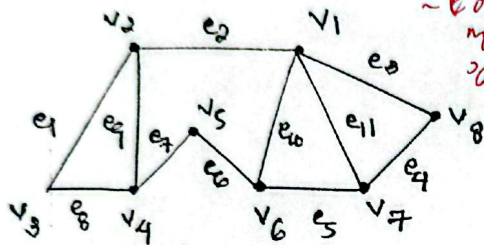
$\delta = \text{degree}$

(i) Determine whether the following graph  $G$  has an Euler cycle.

If the graph  $G$  has an Euler cycle, exhibit one.

Euler cycle  
vertex and degree  
is even (both)  
~ Edge and  
is not  
original

(i)



$$\begin{aligned} \delta(v_1) &= 4, \delta(v_2) = 3, \delta(v_3) = 2 \\ \delta(v_4) &= 3, \delta(v_5) = 2, \delta(v_6) = 3 \\ \delta(v_7) &= 3, \delta(v_8) = 2, \delta(v_9) = \end{aligned}$$

even number  
degree 600

Since the degree of vertices

$v_2, v_5$  and  $v_6$  and  $v_7$  are odd degree.  
So,  $G$  has not an Euler cycle.

(iii) Six edges, eight vertices

For any connected tree with  $v$  vertices, the number of edge  $e$

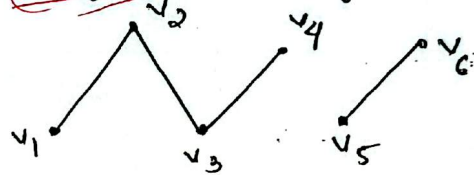
$$e = v - 1.$$

For  $v = 8$ , a connected tree requires  $e = 8 - 1 = 7$  edges.

So, a connected tree does not exist.

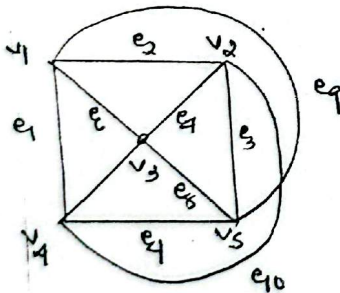
cycle is not possible

A cyclic, Four edges; six vertices



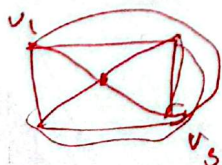
vertices and edges  
edge counting

(ii)



$$\delta(v_1) = \delta(v_2) = \delta(v_3) = \delta(v_4) = \delta(v_5) = 4$$

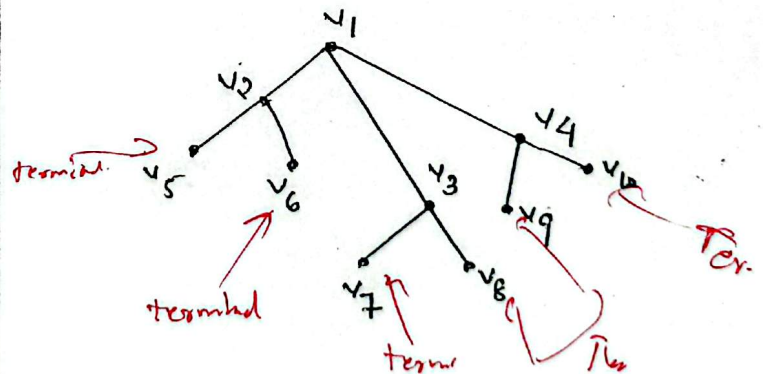
Since the degree of every vertex is even,  $G$  has an Euler cycle. The Euler cycle is



2

$v_1, v_2, v_5, v_4, v_3, v_1, v_4, v_2, v_3, v_5, v_1$

(v) Tree; Four internal vertices; six terminal vertices



all graph are Terminal

```

graph TD
    e((e)) --- a((a))
    e --- b((b))
    e --- f((f))
    b --- c((c))
    b --- d((d))
    d --- g((g))
    d --- h((h))
    g --- i((i))
    g --- j((j))
  
```

A tree diagram  $T'$  with root 'e'. The root 'e' has children 'a' and 'c'. Node 'a' has children 'd' and 'f'. Node 'c' has children 'g' and 'i'. Node 'g' has children 'h' and 'j'. A red line is drawn through the tree, and a red arc is at the bottom.

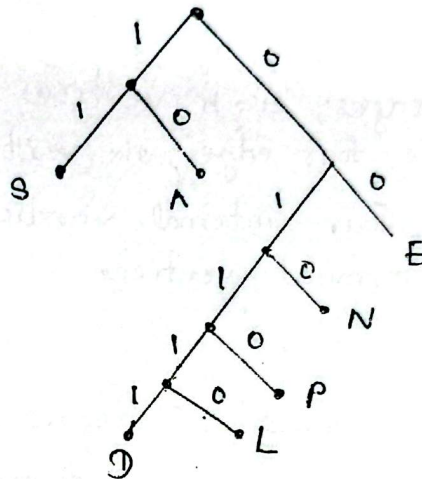
~~2, 1, 2, 3, 2, 3, 3, 3, 4, 4, 5~~

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(i) 01110100110

(ii) 01111 00100110

(iii) 1110011101001111



9012

(i)  $01110 = L$

$$w = A$$
$$0110 = P$$

$\therefore 01110100110 = \text{LAP}$

(ii)  $01111 = D$

$$00 = \mathbb{E}$$
$$l_0 = A$$
$$01110 = L$$

• 011100100110  
= DEAL

(iii)  $n = 8$

 $10 \rightarrow A$ 
$$01110 = L$$

to A

$$0111 = \mathbb{D}$$

• 111001110100111 = SALAD

Draw a graph having the given properties or explain why no such graph exists

(i) Tree; all vertices of degree 2.

(ii) Tree; six vertices having

degrees 1, 1, 1, 1, 3, 3.

(iii) Six edges; eight vertices

(iv) Acyclic; four edges; six vertices

(v) Tree, four internal vertices, six terminal vertices.

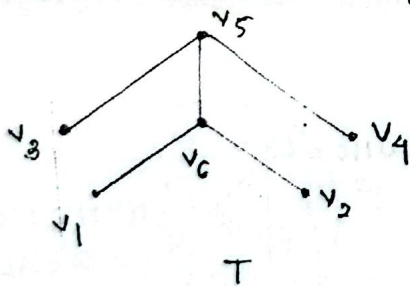
solution,

(i) Tree; all vertices of degree 2.

A terminal vertex has degree 1.

No such graph.

(ii) Tree; six vertices having 1, 1, 1, 1, 3, 3.



(iii)

$$2e \geq 4f$$

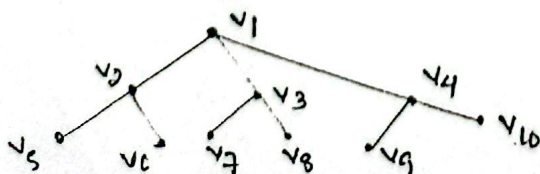
$$2e \geq 4(e-v+2)$$

$$2(9) \geq 4(9-6+2)$$

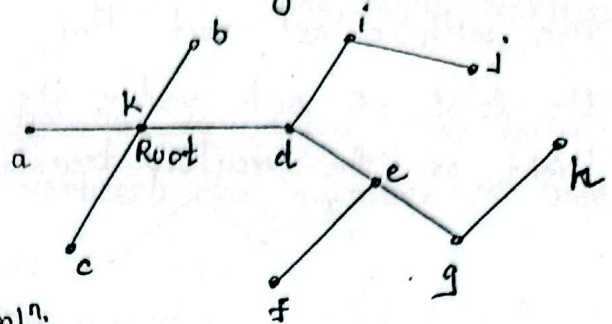
$$18 \geq 4(5)$$

$$18 \geq 20$$

(v) Tree, four internal vertices, six terminal vertices



Find the level of each vertex and the height of the tree.



sol<sup>n</sup>:

The vertices a, b, c, d, e, f, g, h, i, j, k are on (respectively) levels 1, 1, 1, 1, 2, 3, 3, 4, 2, 3, 0. The height of T is 4.

③ Show that the graph  $K_{3,3}$  is not planar.

Proof: Suppose that  $K_{3,3}$  is planar.

Since every cycle has at least four edges, each face is bounded by at least four edges. Thus the number of edges that bound faces is at least  $4f$ . In a planar graph, each edge belongs to at most two bounding cycles. Therefore,

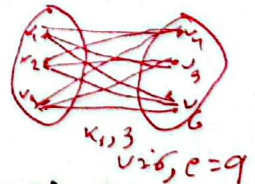
$$2e \geq 4f$$

$$2e \geq 4(e-v+2)$$

$$e=9, v=6, 18 \geq 20$$

$$9 \geq 4(9-6+2) = 20$$

which is contradiction. Therefore,  $K_{3,3}$  is not planar.



$$2e \geq 4(e-v+2)$$

(a) solution,

| path          | length        |
|---------------|---------------|
| a, b, c, d, e | $8+6+3+4=21$  |
| a, b, d, c, e | $8+12+3+5=28$ |
| a, c, b, d, e | $2+6+12+4=24$ |
| a, c, d, b, e | $2+3+12+9=26$ |
| a, d, b, c, e | $4+12+6+5=27$ |
| a, d, e, b, c | $4+3+6+9=22$  |

The path of minimum length from a to e is a, b, c, d, e with length of 21.

(b)  $v=c, w=d$

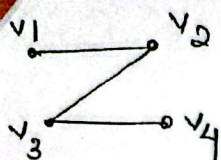
| path          | length        |
|---------------|---------------|
| c, a, b, e, d | $2+8+9+4=23$  |
| c, a, e, b, d | $2+6+9+12=29$ |
| c, b, a, e, d | $5+8+6+4=23$  |
| c, b, e, a, d | $6+9+6+4=25$  |
| c, e, a, b, d | $5+6+8+12=31$ |
| c, e, b, a, d | $5+9+8+4=26$  |

The path of minimum length from c to d is c, a, b, e, d with length of 23.

(c)  $v=a, w=b$

| path          | length |
|---------------|--------|
| a, c, d, e, b |        |
| a, d, e, b    |        |
| a, e, c, d, b |        |
| a, c, e, d, b |        |
| a, c, d, e, b |        |
| a, d, e, c, b |        |

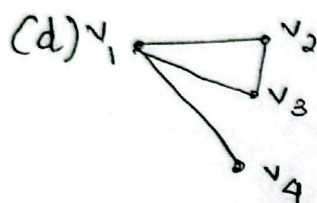
The path of minimum length from a to b is with length of .



The graph  $G$  is bipartite graph.

The disjoint vertex sets  $V_1 = \{v_1, v_3\}$

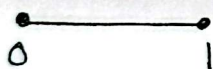
and  $V_2 = \{v_2, v_4\}$ .



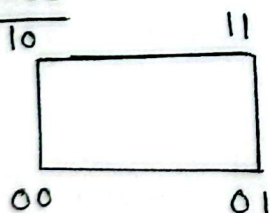
The graph  $G$  is not bipartite graph.

### ⑤ The $n$ -cube (Hypercube)

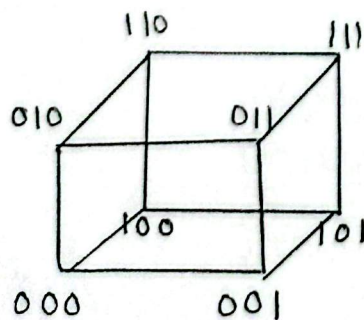
1 cube



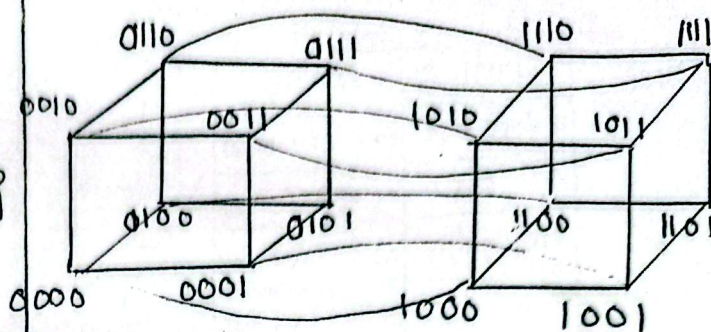
2 cube



3 cube



$n$  cube



The number of vertices in  $n$ -cube  $= 2^n$  vertices.

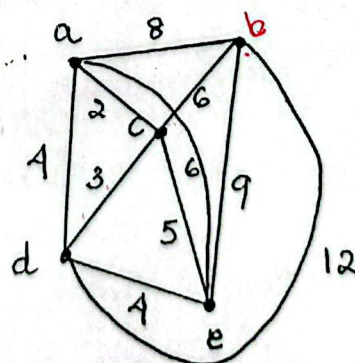
The number of edges in  $n$ -cube  $= n \cdot 2^{n-1}$  edges

⑥ Find a path of minimum length from  $v$  to  $w$  in the graph of figure that passes through each vertex exactly one time.

(i)  $v = a, w = e$

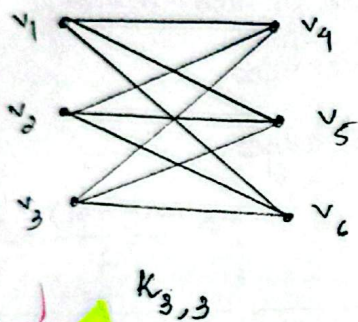
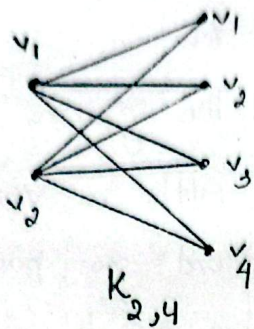
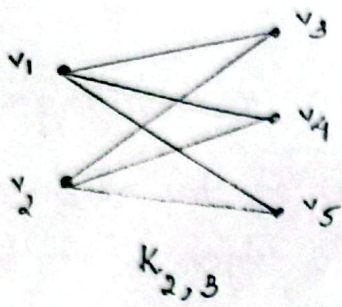
(ii)  $v = c, w = d$

(iii)  $v = a, w = b$ .



For  $K_{m,n}$  (Bipartite G)

2) Draw  $K_{2,3}$ ,  $K_{2,4}$  and  $K_{3,3}$ .



3) Find a formula for the number of  $K_{m,n}$ .

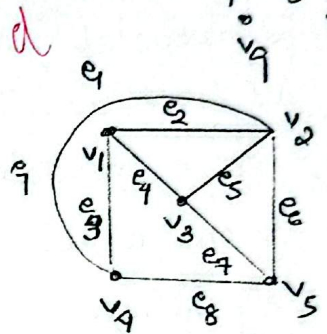
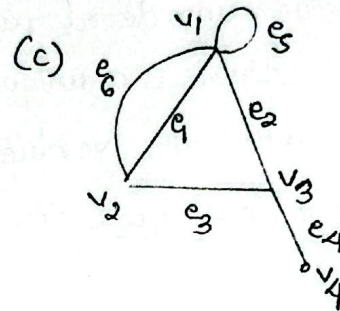
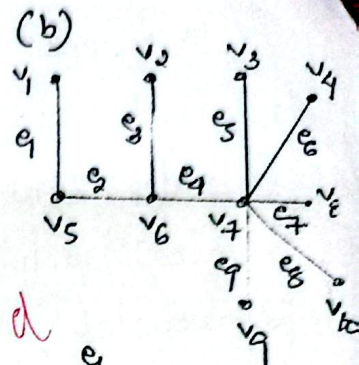
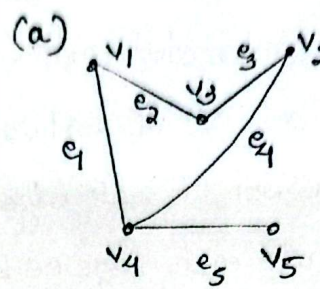
solution,

By definition, a complete bipartite graph  $K_{m,n}$  divides its vertices into two disjoint sets,  $V_1$  and  $V_2$ .

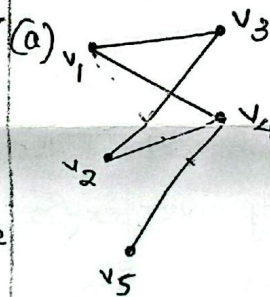
The vertex set  $V_1$  contains  $m$  vertices and the vertex set  $V_2$  contains  $n$  vertices.

The total number of edges in  $K_{m,n}$  is  $m \times n = mn$  edges.

4) If the graph is bipartite specify the disjoint vertex sets.



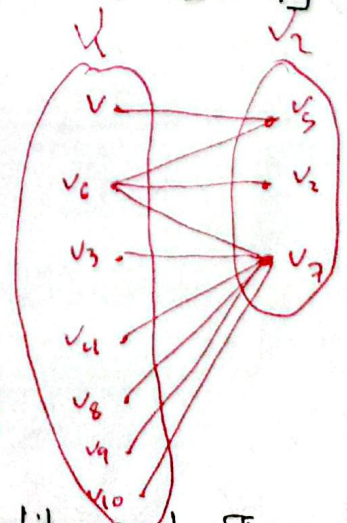
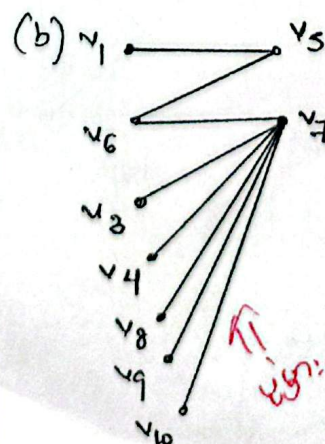
solution



The graph is bipartite graph.

The disjoint vertex sets

$V_1 = \{v_1, v_2, v_5\}$  and  $V_2 = \{v_3, v_4\}$ .



The graph is bipartite graph. The disjoint vertex sets  $V_1 = \{v_1, v_3, v_4, v_6, v_8, v_9, v_{10}\}$ ,  $V_2 = \{v_5, v_7\}$ .

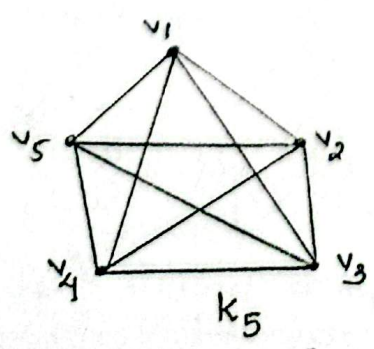
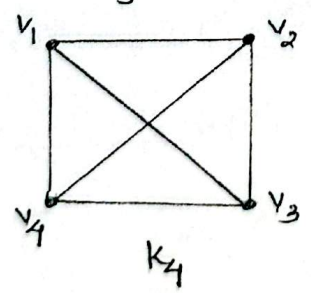
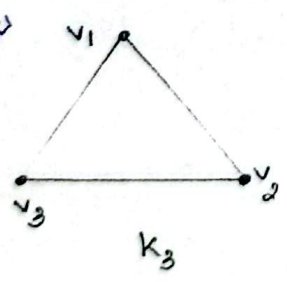
Definition

A graph (or undirected graph)  $G$  consists of a set  $V$  of vertices (or nodes) and a set  $E$  of edges (or arcs) such that each edge  $e \in E$  is associated with an unordered pair of vertices. If there is a unique edge  $e$  associated with the vertices  $v$  and  $w$ , we write  $e = (v, w)$  or  $e = (w, v)$ .

Definition

The graph  $K_n$  called the complete graph on  $n$  vertices, has  $n$  vertices and every vertex is joined to every other vertex by an edge.

Q1 Draw  $K_3$ ,  $K_4$  and  $K_5$ .



$$5C_2 = \frac{5 \times 4}{1 \times 2} = 10$$

2104 + 2110  
Q2 Find a formula for the number of edges in  $K_n$ .  
solution,

The graph  $K_n$  is a complete graph with  $n$  vertices. It has no loop and no parallel edges. So it is a simple graph. There is at most one edge between ~~two distinct~~ two distinct vertices.

The number of edges =  $nC_2$   
=  $\frac{n(n-1)}{2}$  edges

Definition

A graph in which the vertices can be partitioned into disjoint sets  $V_1$  and  $V_2$  with every edge incident on one vertex in  $V_1$  and one vertex in  $V_2$  is called a bipartite graph.

The graph  $K_{m,n}$  called the complete bipartite graph on  $m$  and  $n$  vertices has disjoint sets  $V_1$  of  $m$  vertices and  $V_2$  of  $n$  vertices. Every vertex in  $V_1$  is joined to every vertex in  $V_2$  by an edge.